

Project Management

CPM An Illustrative Example

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The House Construction Problem

The Build-Rite Construction Company has identified ten activities that take place in building a house. They are

Activity		Immediate Predecessors	Expected Time (days)
1	Walls and Ceiling	2	5
2	Foundation	-	3
3	Roof Timbers	1	2
4	Roof Sheathing	3	3
5	Electrical Wiring	1	4
6	Roof Shingles	4	8
7	Exterior Siding	8	5
8	Windows	1	2
9	Paint	6, 7, 10	2
10	Inside Wall Board	8, 5	3

Critical Path Method (CPM)

Step 1: Forward pass

Step 2: Backward pass

Step 3: Calculating

Identify the critical path(s).

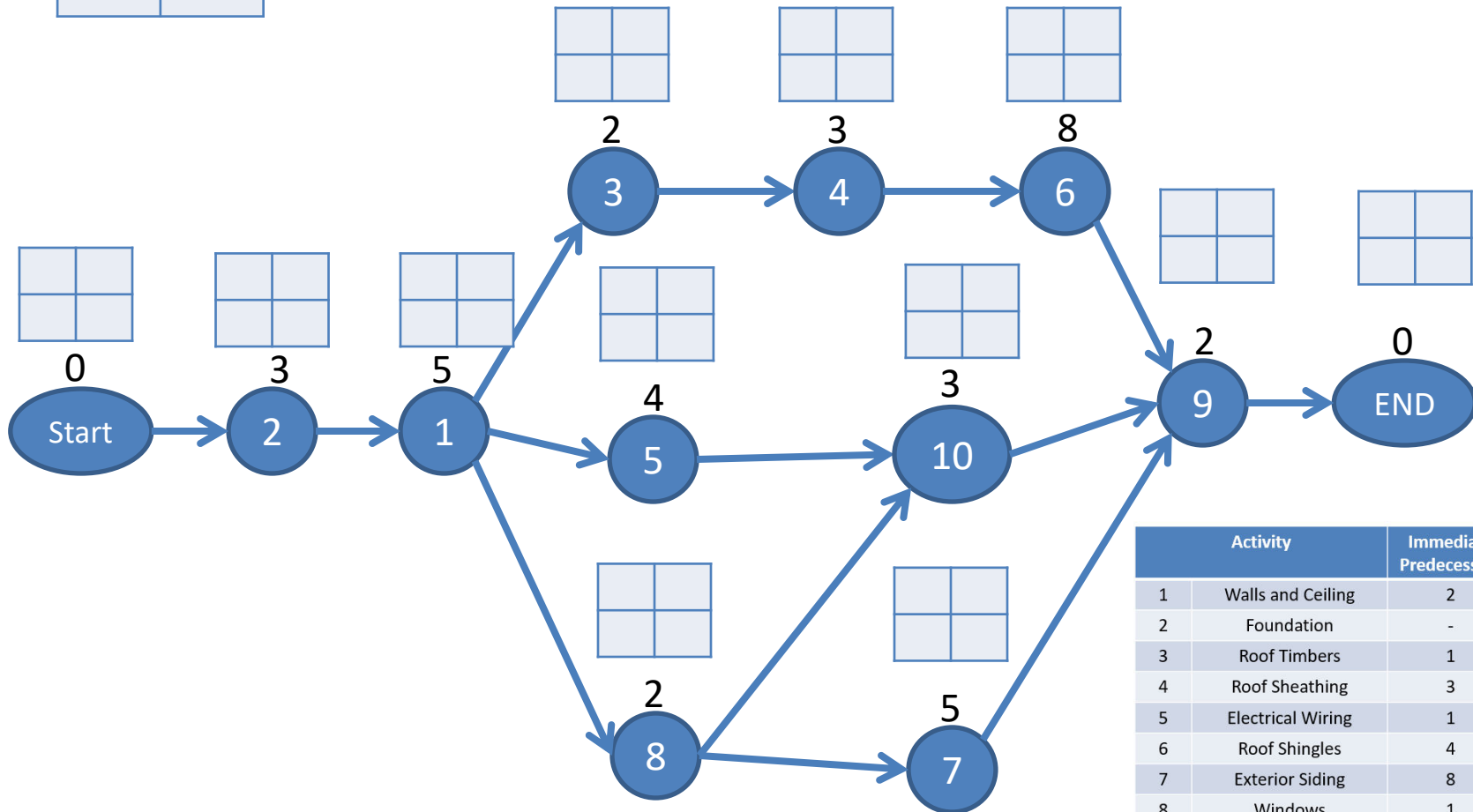
interpret the meaning of slacks and critical activities.

Notations

ES	EF
LS	LF

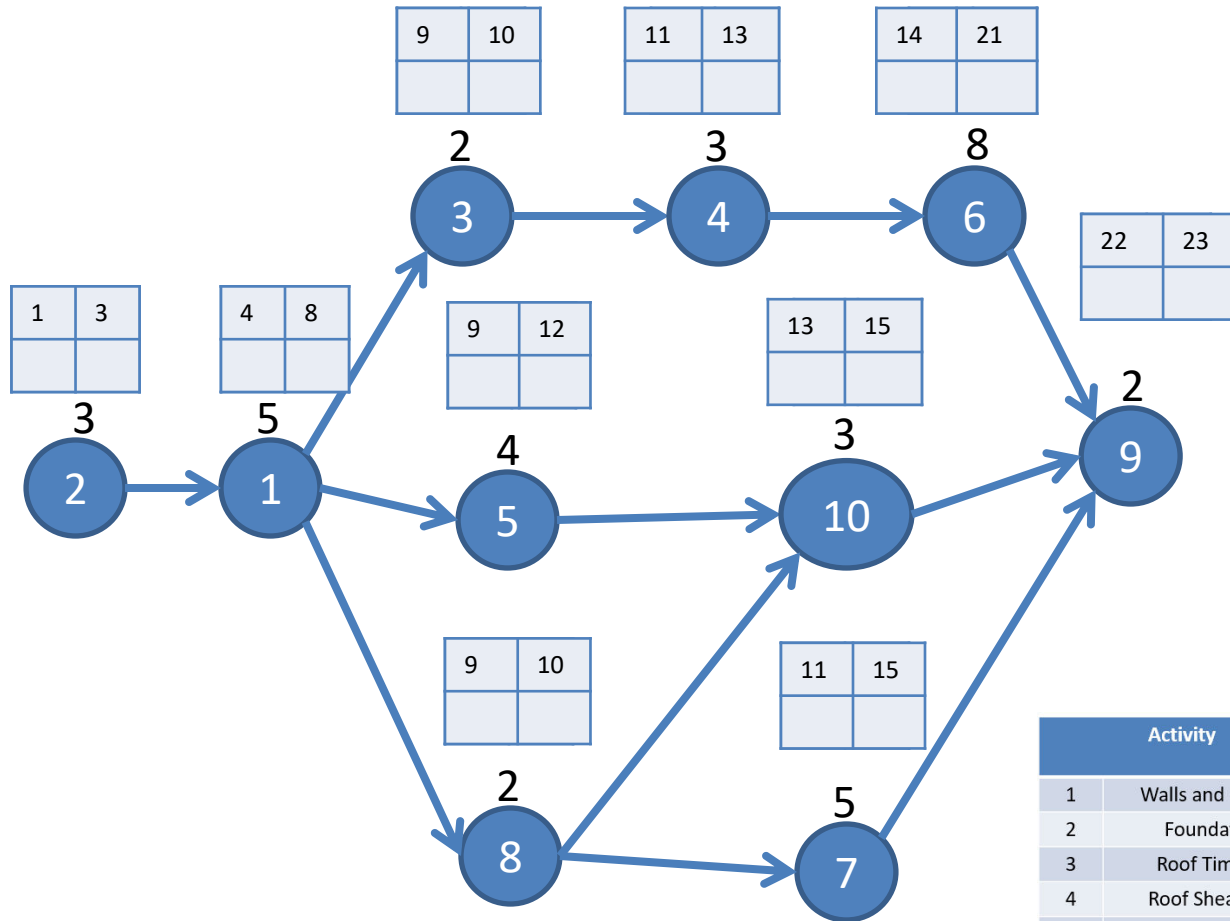
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TS: Total Slack
FS: Free Slack



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Step 1: Forward Pass



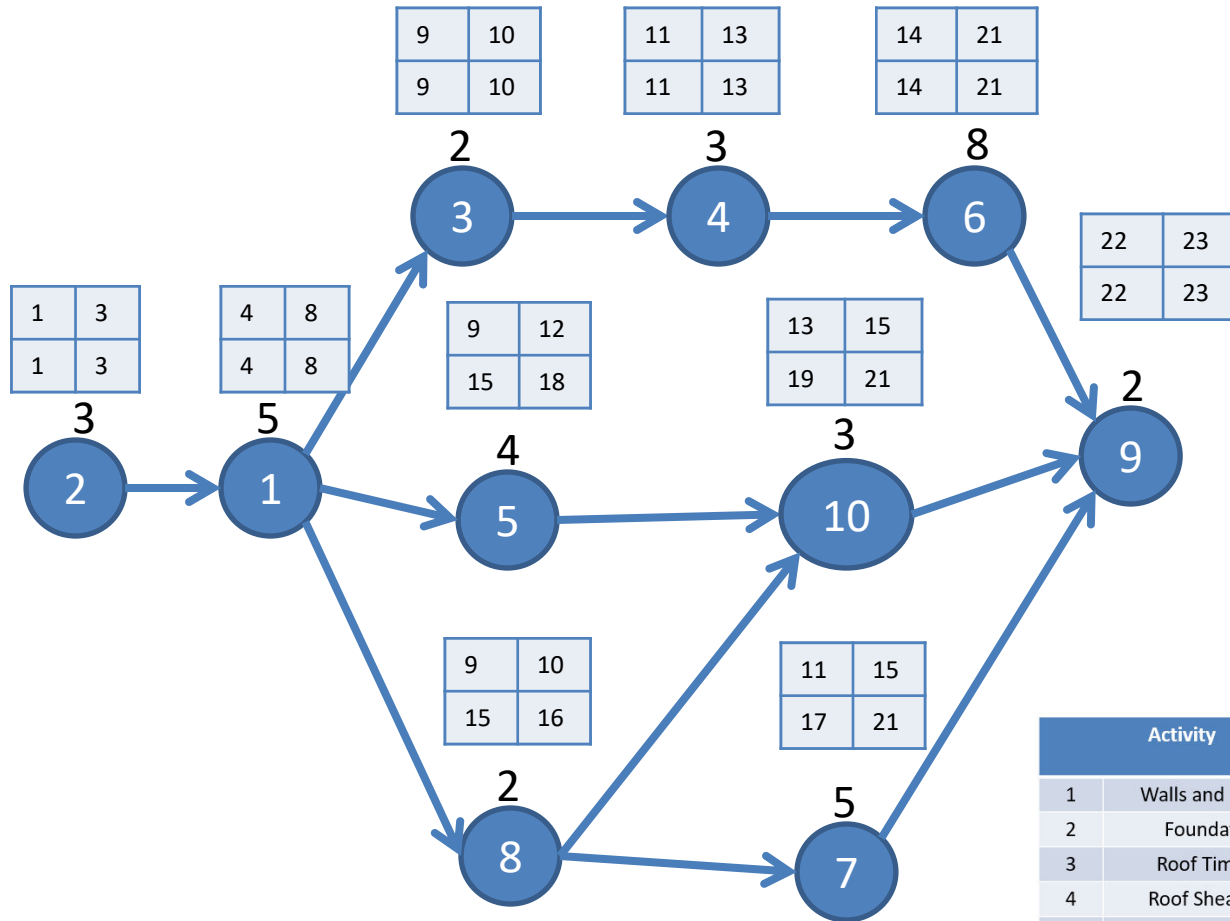
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Step 2: Backward Pass



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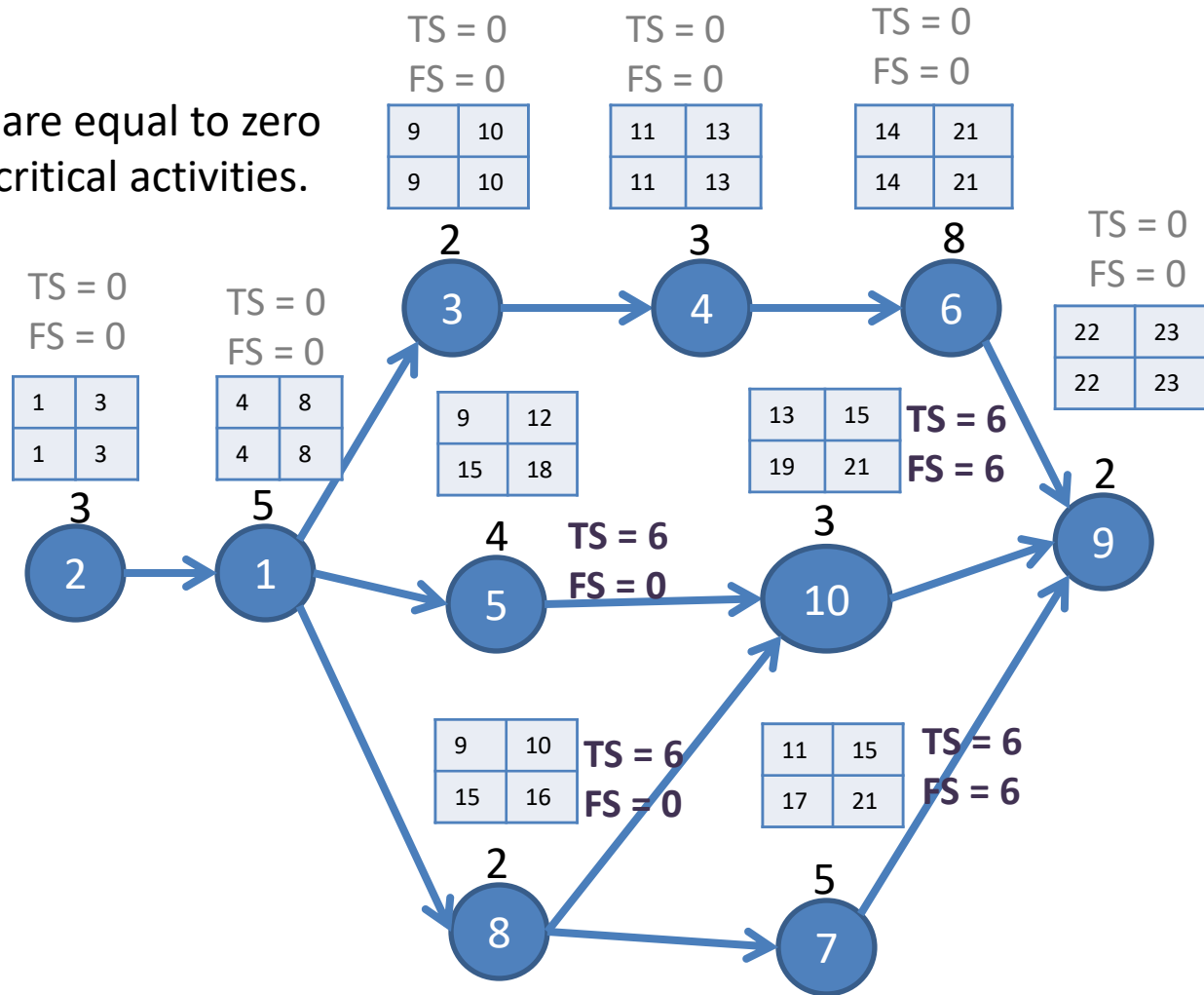
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Step 3: Calculating Slacks

The slacks are equal to zero for all the critical activities.



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